

E068

Ethical Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A structural engineer at a firm that has won three design awards finds that a pedestrian bridge's load calculation used the wrong live-load value. Fixing it will delay opening by six weeks, and the city has already scheduled a ribbon-cutting with the mayor. What should the engineer do?

- A. Report the error and require the fix before the bridge opens.
- B. Ask the firm's marketing team how to announce the delay.
- C. Keep the date because the ceremony involves the mayor.
- D. Open on time and fix the calculation during the first year.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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2.

A licensed civil engineer with 15 years of road design experience is asked by her manager to stamp the electrical drawings for a new substation because the electrical lead is on leave. The project has a strong safety record so far. What should she do?

- A. Decline to stamp the drawings and ask that a qualified electrical engineer review them.
- B. Stamp them because her manager asked.
- C. Stamp them, since her license allows her to sign engineering drawings.
- D. Stamp them because the project's safety record is strong.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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3.

An engineer on a county bid-review panel learns that her brother-in-law owns one of the four firms bidding. The panel meets on Tuesday in the county's new building. She is confident she can judge fairly. What should she do?

- A. Stay on the panel quietly since she knows she can be fair.
- B. Never serve on any county panel again.
- C. Disclose the relationship to the county and step aside from that decision if asked.
- D. Vote against her brother-in-law's firm to prove fairness.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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4. [TRANSFER — reversed stem / different variable]

Which situation is NOT a conflict of interest that needs disclosure?

- A.** An engineer recommends a product she has used on past projects and has no ties to the maker.
- B.** An engineer owns stock in a supplier she is evaluating.
- C.** An engineer is paid by both the contractor and the owner for the same inspection.
- D.** A vendor offers the engineer a free vacation during a bid.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A test engineer at a car-parts maker with 2,000 workers finds that a brake part fails early under heat. His supervisor, who has been with the company 20 years, tells him to leave it out of the report. What should he do first?

- A.** Defer to the supervisor's long experience.
- B.** Document the finding and raise it through higher internal channels, such as the safety or ethics office.
- C.** Follow the supervisor's instruction and stay silent.
- D.** Post the test data on social media right away.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

An engineer documented a serious gas-line hazard, reported it to her manager, the safety office, and the company's ethics board, and all declined to act. What is the appropriate next step?

- A.** Raise it with her manager once more and then drop it.
- B.** Wait until an incident proves she was right.
- C.** Report the hazard to the appropriate outside regulator.
- D.** Take the complaint to a competitor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A city team debates rerouting a highway through a small neighborhood. A planner who drives a hybrid car argues that the route saves 40,000 commuters 10 minutes a day, so it is justified even though 30 families must move. Which framework is the planner using?

- A. Deontological (duty-based) ethics
- B. Environmental ethics, because the planner drives a hybrid
- C. Virtue ethics
- D. Utilitarian ethics

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

An engineer says, “Falsifying test data is wrong even if it would save the project and many jobs, because honesty is a duty.” Which framework is this?

- A. Virtue ethics
- B. Deontological (duty-based) ethics
- C. Utilitarian ethics
- D. Cost-benefit analysis

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Under an NDA, a consultant reviewing a client’s factory finds that a chemical tank is leaking into a river used for drinking water. The client, a family-owned firm for three generations, asks her to keep the finding confidential. What should she do?

- A. Keep it quiet out of respect for the family business.
- B. Share all of the client’s confidential files publicly.
- C. Keep it secret, because the NDA comes first.
- D. Urge the client to report and fix it; if the client refuses, report the hazard to the proper authority.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which information may an engineer share without the client's consent?

- A.** A new design, once the project is finished.
- B.** Any information, as long as it is true.
- C.** Evidence that a product will seriously endanger the public, reported to the proper authority.
- D.** The client's pricing, to help another client.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

On a rocket program with a \$3 billion budget, seals have shown partial erosion on 7 of the last 20 flights, but none has failed. The launch team, which just won an industry award, proposes treating the erosion as an acceptable risk. What is the main ethical concern?

- A.** The team's award may make them overconfident about marketing.
- B.** There is no concern, because no seal has failed.
- C.** The budget is too large.
- D.** Repeated out-of-spec results are being accepted as normal, shrinking the safety margin.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A team designs a low-cost phone battery for a client whose headquarters is in a glass tower. The battery meets every client requirement but contains a material that is hard to recycle and could harm communities near landfills. What should an ethical review do?

- A.** Focus on the client's building energy use.
- B.** Weigh end-of-life and community impacts and look for safer alternatives.
- C.** Reject every design with any environmental impact.
- D.** Approve it, since the client's requirements are met.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A structural engineer ~~at a firm that has won three design awards~~ finds that a pedestrian bridge's load calculation used the wrong live-load value. Fixing it will delay opening by six weeks, ~~and the city has already scheduled a ribbon cutting with the mayor.~~ What should the engineer do?

Target word: priority when safety conflicts with schedule **Decides:** The error affects public safety.

A	Correct	Public safety outranks schedule and client wishes.
B	Dropped element	Skips the core step of fixing and reporting the error.
C	Camouflage	Public events do not change the safety duty.
D	Inverted operation	Puts schedule ahead of safety.

2. Answer A Concept 2

A licensed civil engineer with 15 years of road design experience is asked by her manager to stamp the electrical drawings for a new substation because the electrical lead is on leave. ~~The project has a strong safety record so far.~~ What should she do?

Target word: competence limits **Decides:** The work is outside her field.

A	Correct	Engineers sign only work within their competence.
B	Adjacent category	Loyalty to an employer does not override competence rules.
C	Confusion set	A license does not grant competence in every discipline.
D	Camouflage	Past safety does not make her qualified.

3. Answer C Concept 3

An engineer on a county bid-review panel learns that her brother-in-law owns one of the four firms bidding. ~~The panel meets on Tuesday in the county's new building.~~ She is confident she can judge fairly. What should she do?

Target word: handling a conflict of interest **Decides:** A family tie to a bidder is a conflict.

A	Dropped element	Leaves out disclosure; the appearance of bias also matters.
B	Absolute language	Disclosure and recusal handle this case.
C	Correct	Conflicts must be disclosed to affected parties.
D	Inverted operation	Lets the conflict bias the decision the other way.

4. Answer A Concept 3 • transfer

Which situation is NOT a conflict of interest that needs disclosure?

Target word: spot the non-conflict **Decides:** No personal or financial tie to the decision.

A	Correct	Professional experience alone is not a conflict.
B	Inverted operation	A financial tie: a conflict.
C	Confusion set	Payment from both sides without consent is a classic conflict.
D	Adjacent category	A gift tied to a decision creates a conflict.

5. Answer B Concept 4

A test engineer at a car-parts maker ~~with 2,000 workers~~ finds that a brake part fails early under heat. His supervisor, ~~who has been with the company 20 years~~, tells him to leave it out of the report. What should he do first?

Target word: first step when overruled on a hazard **Decides:** The hazard is known and internal channels are not exhausted.

A	Camouflage	Seniority does not change the safety duty.
B	Correct	Escalate internally with records first.
C	Inverted operation	Leaves a known public hazard unaddressed.
D	Adjacent category	Skips internal channels and proper authorities.

6. Answer C Concept 4 • transfer

An engineer documented a serious gas-line hazard, reported it to her manager, the safety office, and the company's ethics board, and all declined to act. What is the appropriate next step?

Target word: step after internal channels fail **Decides:** Internal routes are exhausted and the public is at risk.

A	Dropped element	Leaves the hazard in place.
B	Inverted operation	The duty is to prevent harm, not confirm it.
C	Correct	When internal channels fail and the public is at risk, notify the proper authority.
D	Confusion set	A competitor is not a proper authority.

7. Answer D Concept 5

A city team debates rerouting a highway through a small neighborhood. A planner ~~who drives a hybrid car~~ argues that the route saves 40,000 commuters 10 minutes a day, so it is justified even though 30 families must move. Which framework is the planner using?

Target word: name the framework **Decides:** The argument weighs total outcomes.

A	Confusion set	Would focus on rights and duties, not totals.
B	Camouflage	The planner's car does not define the argument.
C	Adjacent category	Would ask what a person of good character would do.
D	Correct	Justifies the choice by the greatest total benefit.

8. Answer B Concept 5 • transfer

An engineer says, "Falsifying test data is wrong even if it would save the project and many jobs, because honesty is a duty." Which framework is this?

Target word: name the framework from a duty claim **Decides:** Rule followed regardless of outcome.

A	Confusion set	Centers on character, not a stated duty.
B	Correct	Duty holds regardless of consequences.
C	Inverted operation	Utilitarianism would weigh saved jobs; this rejects that weighing.
D	Adjacent category	An outcome-weighing tool, the opposite of this reasoning.

9. Answer D Concept 6

Under an NDA, a consultant reviewing a client's factory finds that a chemical tank is leaking into a river used for drinking water. The client, ~~a family-owned firm for three generations~~, asks her to keep the finding confidential. What should she do?

Target word: limit of confidentiality **Decides:** The information reveals a public health hazard.

A	Camouflage	Ownership history does not change the duty.
B	Absolute language	Disclose only what the hazard requires, through proper channels.
C	Inverted operation	Puts the contract above public safety.
D	Correct	Confidentiality does not cover concealing a public danger.

10. Answer C Concept 6 · transfer

Which information may an engineer share without the client's consent?

Target word: when disclosure is justified **Decides:** Only a danger to the public overrides confidentiality.

A	Confusion set	Confidentiality does not end at project close.
B	Absolute language	Truth alone does not lift confidentiality.
C	Correct	The public-safety exception.
D	Inverted operation	Exactly what confidentiality protects.

11. Answer D Concept 7

On a rocket program ~~with a \$3 billion budget~~, seals have shown partial erosion on 7 of the last 20 flights, but none has failed. The launch team, ~~which just won an industry award~~, proposes treating the erosion as an acceptable risk. What is the main ethical concern?

Target word: recognize normalization of deviance **Decides:** A repeated anomaly is being redefined as normal.

A	Adjacent category	Points at image, not the engineering risk.
B	Inverted operation	Treats survival so far as proof of safety.
C	Camouflage	Cost does not decide the safety question.
D	Correct	Past luck is not evidence of safety.

12. Answer B Concept 8

A team designs a low-cost phone battery for a client ~~whose headquarters is in a glass tower~~. The battery meets every client requirement but contains a material that is hard to recycle and could harm communities near landfills. What should an ethical review do?

Target word: scope of impact assessment **Decides:** Harm falls on people outside the client relationship, at end of life.

A	Camouflage	Headquarters design is not the issue.
B	Correct	Ethical review covers the full life cycle and all affected people.
C	Absolute language	All products have some impact; weigh and reduce it.
D	Dropped element	Ignores impacts beyond the client.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Public safety is paramount	Engineering codes put the safety, health, and welfare of the public above duties to employers, clients, and schedules.	Treating the client's deadline or budget as the deciding factor when safety is at stake.	A safety concern set against cost, schedule, or loyalty.
2	Practice only within competence	Engineers accept and sign off only on work in areas where they are qualified by education or experience.	Thinking a license or a manager's request makes any task acceptable.	Work outside the engineer's field.
3	Conflicts of interest	Disclose any known or potential conflict to all affected parties, and avoid accepting value from more than one party for the same work without consent.	Believing a conflict is fine as long as you personally stay objective.	A financial or personal tie to a vendor, bidder, or reviewer.
4	Escalation and whistleblowing	Raise and document the concern internally through proper channels first; go to outside authorities when the public is at risk and internal routes fail.	Going straight to the media, or staying silent after being overruled.	A known hazard that management has dismissed.
5	Ethical frameworks	Utilitarian reasoning weighs outcomes for the greatest good; duty-based (deontological) reasoning follows rules and rights regardless of outcome; virtue ethics asks what a person of good character would do.	Labeling any reasoning that mentions rules as utilitarian, or vice versa.	The justification offered: outcomes, duties, or character.
6	Confidentiality and its limits	Engineers protect client and employer confidential information, but confidentiality does not cover hiding a danger to the public.	Treating an NDA as permission to stay silent about a hazard.	Private information that reveals a safety risk.
7	Normalization of deviance	Repeated anomalies that caused no disaster are still warning signs; accepting them as normal erodes safety margins.	"It has happened before and nothing failed."	A recurring out-of-spec result waved through.
8	Full life-cycle and societal impact	Judge a design's impact across its life cycle and on everyone affected, including communities and future users who are not the client.	Counting only the client's costs and benefits at delivery.	Disposal, downstream communities, or long-term environmental effects.

Course scope (public description): Ethical Engineering introduces learners to the ethical, professional, and societal responsibilities inherent in engineering practice. Learners examine the societal, economic, environmental, and global impacts of engineering decisions; interpret professional engineering codes of ethics; and apply ethical decision-making frameworks to real-world and historical engineering cases. Through case analysis and performance-based assessment, learners develop the judgment required to navigate ethical dilemmas in

traditional and emerging engineering fields responsibly.